臺大醫院　腫瘤醫學部　放射腫瘤科

102 年放射治療物理學期中考核(答案版)

姓名:__________　　　　　　　　　　　　　　　　　　分數:________

(50 題，每題 2 分)

C　1.Which statement about isodose curve is <u>not true</u>?

(A)Lines passing through points of equal dose.

(B)Drawn at regular intervals of absorbed dose.

(C)Expressed as a percentage of the absorbed dose at any depth *d* to a reference depth d_0.

(D)Field size determined on the basis of isodose curve (e.g. 90%) adequately cover the target volume.

D　2.Which statement about penumbra region is <u>not true</u>?

(A)Dose rate decreases rapidly as a function of lateral distance from the beam axis.

(B)Geometric penumbra width depends on source size, distance from the source, and source-to-diaphragm distance.

(C) Physical penumbra width: lateral distance between two specified isodose curves at a specified depth (e.g. lateral distance between 90% and 20% isodose line at D max).

(D)Geometric penumbra is the best measure of the beam sharpness near the edges.

C　3.Which statement is <u>not true</u>?

(A)The depth of a given isodose curve increases with beam quality

(B)Greater lateral scatter associated with lower-energy beams causes isodose curves outside field to bulge out

(C)When using parallel opposing fields, the central axis maximum dose near the surface increases relative to the midpoint dose with the beam energy increase.

(D)For megavoltage beams, the scatter outside the field is minimized as a result of forward scattering and becomes more a function of collimation than energy

B 4.Which statement about wedge is <u>not true</u>?

(A)Made of a dense material, such as lead or steel.

(B)Arranged at a distance of at least 15 cm from the skin surface to destroy skin-sparing effects.

(C)Wedge angle, the angle through which an isodose curve is titled at the central ray of a beam at a specified depth, measured between the isodose curve and the normal to the central axis.

(D)Wedge factor is the ratio of doses with and without the wedge, at a point in phantom along the CAX of the beam.

D 5.Rotation therapy is

(A) Suitable for treatment volume is too large.

(B) Suitable for the external surface differs markedly from a cylinder.

(C) Suitable for the tumor is too far off center.

(D) Suitable for small and deep-seated tumors.

請寫出以下縮寫全名:

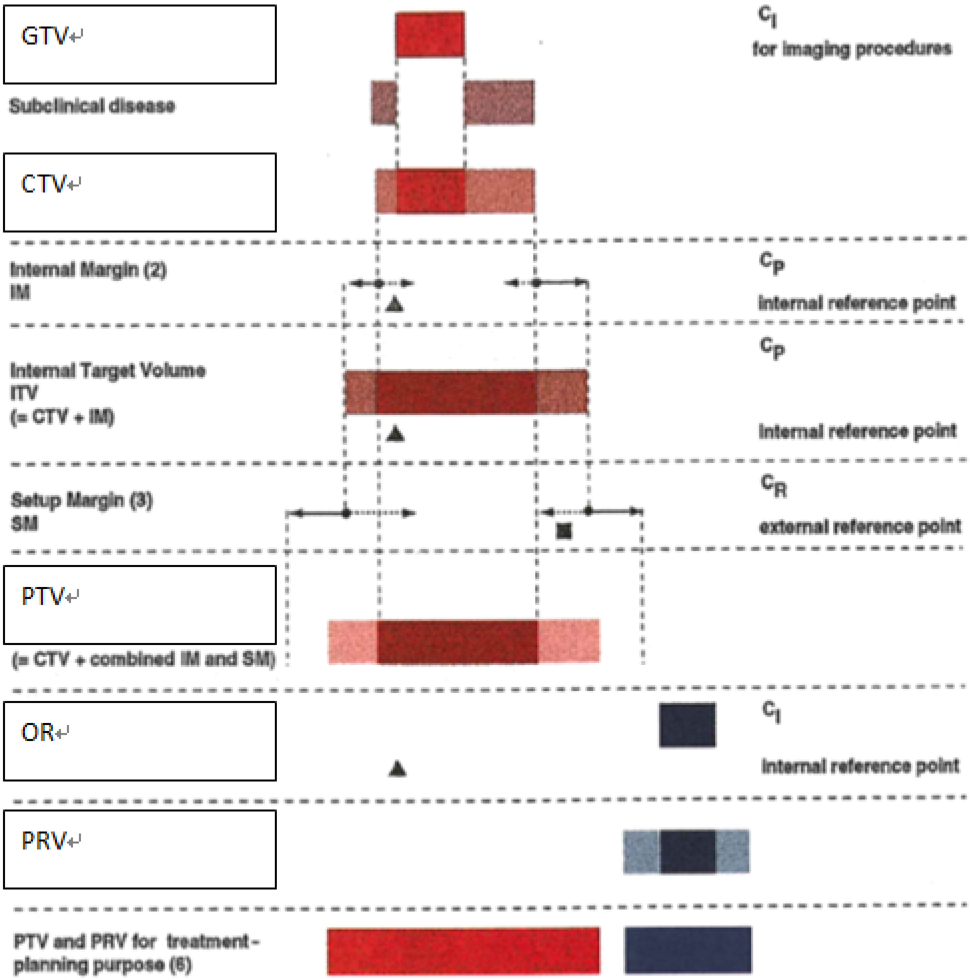
6.GTV <u>Gross tumor volume</u>

7.CTV <u>Clinical target volume</u>

8.PTV <u>Planning target volume</u>

9.OR <u>Organ at risk</u>

10.PRV <u>Planning organ at risk volume</u>



B 11.放射治療在勾畫靶區時，常參考 CT 和 MRI 影像，下列有關 CT 和 MRI 的比較何者為非？

A. MRI 提供較好的軟組織比對

B. MRI 提供較高的空間解析度

C. CT 孔徑較大，臨床上放射治療使用 Big bore CT 可容許治療所需較寬的擺位姿勢

D. MRI 可直接擷取 axial, sagittal, coronal 等軸向切面影像

A 12.下列有關傳統模擬定位機敘述何者有誤？

A. 使用 KV 級 X 光，光與物質主要作用為康普吞效應

B. 與直線加速器擁有同樣的幾何、機械和光學等特性

C. 同時具備透視和照相功能

D. 所拍攝之定位片(SIM film)較直線加速器驗證片(port film)影像清晰

C　13.有關電子照野驗證片(EPID)描述何者為非？

A.　提供即時的影像

B.　使用電子式成像技術，解決傳統底片存放問題

C.　使用 KV 級 X 光，故影像較底片式清晰

D.　如治療前拍攝並確認位置，可視為 IGRT 的一種方式

D　14.下列有關不規則身體輪廓的劑量計算修正何者為非？

A.　TAT(TMR) method 最為準確，治療計劃系統常採用此法

B.　Isodose shift method 為一種經驗法則

C.　高能量(10MV)光子射束使用 Generalized Batho method 計算組織中的不均質。

D.　不論使用何者方法，在所有臨床照射狀況下誤差皆可小於 5%。

D　15.利用手算劑量忽略了射束中組織不均勻的修正時,下列何種情況會使位於該不均勻組織後之給予劑量，低估的情形最嚴重？

A.　使用 4 MV 光子經過肺組織

B.　使用 6 MV 光子經過肺組織

C.　使用 10 MV 光子經過肺組織

D.　使用鈷六十光子經過肺組織

B　16.已知血塊的直線衰減係數為水的 1.2 倍，請問 CT 值約為何？

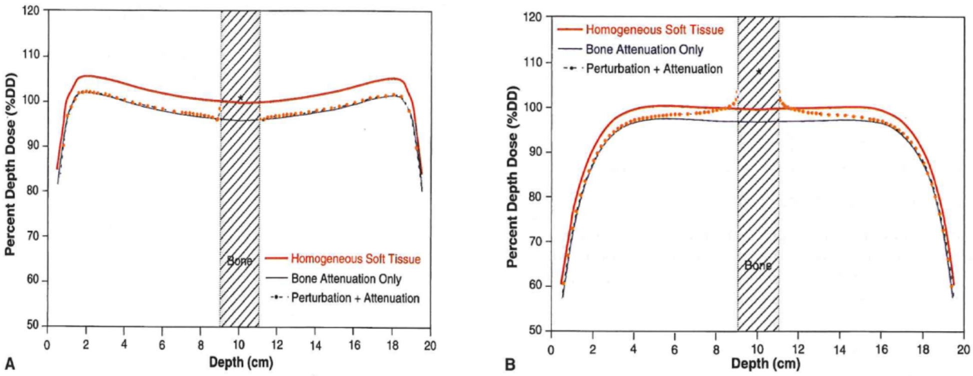
A.　100

B.　200

C.　300

D.　500

B　17.有關光子兩側相對射束(parallel opposed beams)，圖(A)與圖(B)敘述下列何者有誤？



A.　劑量減少由於骨衰減所造成的屏蔽效應

B.　劑量減少的量與骨密度無關

C. 圖(B)是高能量光子，主要作用是成對發生。

D. 以 24 MV 光子兩側對照為例，在穿過 1 cm 厚骨頭，骨與軟組織介面劑量增加約 5%。

C 18.以 10 MV 光子為例，有關組織不均勻的修正下列何者有誤？

A. 射束穿過 1cm 健康的肺，在鄰近肺部後方的組織劑量大約增加 2%。

B. 射束穿過 1cm 的骨頭，在鄰近骨頭後方的組織劑量大約減少 2%。

C. 在骨頭內的軟組織，劑量大約減少 5%。

D. 骨頭周圍的軟組織之吸收劑量有增強的變化。

B 19.利用高能量的光子射束做治療時，常常會用組織補償器（compensator）來改善劑量分布曲線，下列關於組織補償器的敘述何者正確？

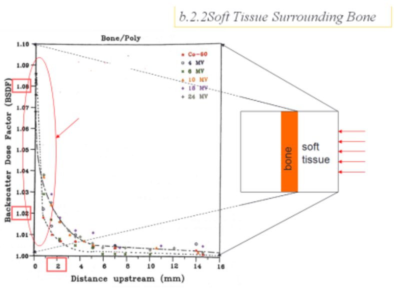
A.放置的位置與 bolus 相同

B.必須放置於離皮膚至少 20 cm 處

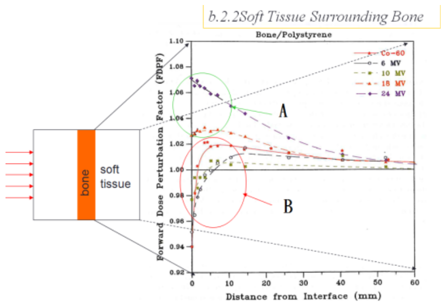
C.較適宜使用低原子序的物質製作

D.不會造成任何光子的衰減

B 20.骨周圍軟組織的吸收劑量變化敘述何者有誤？



(圖 A)



(圖 B)

A. 在光子射束入射端(圖 A)，由於電子回散射，鄰近骨的軟組織吸收劑量有增強的變化。

B. (圖 A)回散射電子射程很短，約 0~2cm 的射程範圍。

C. 在光子射束穿透端(圖 B)，從骨相前散射(Forward scatter)電子造成劑量擾動，依光子能量、距射源的距離而定。

D. (圖 B)中的 A 指的是高能量光子(>10 MV)之擾動。

D 21.A primary beam transmission of a% through the block is considered acceptable for most clinical situations. The b is the number of half-value layers would achieve this transmission.

(A) a=3 , b=3
(B) a=3 , b=5
(C) a=5 , b=3
(D) a=5 , b=5

C 22.In the megavoltage range of photon beams, which one is the most commonly used cerrobend block thickness ?

(A) 5.5 cm

(B) 6.5 cm

(C) 7.5 cm

(D) 8.5 cm

D 23.Which material is the most component in the cerrobend block ?

(A) lead
(B) tin
(C) cadmium
(D) bismuth

C 24.The leaf thickness is sufficient to provide primary x-ray transmission through the leaves of less than **a**% (compared with about **b**% for jaws and **c**% for Cerrobend blocks).

(A) a=1.0, b=2.0, c=3.5
(B) a=3.5, b=1.0, c=2.0
(C) a=2.0, b=1.0, c=3.5
(D) a=3.5, b=2.0, c=1.0

B 25.Which material is the best γ-ray absorbers to reduce the skin dose ?

(A) Low atomic number
(B) Medium atomic number
(C) High atomic number
(D) No relationship

B 26.The skin dose associated with oblique angles is that as the surface dose <u>a</u> with the angle of incidence, the depth of maximum buildup <u>b</u>.

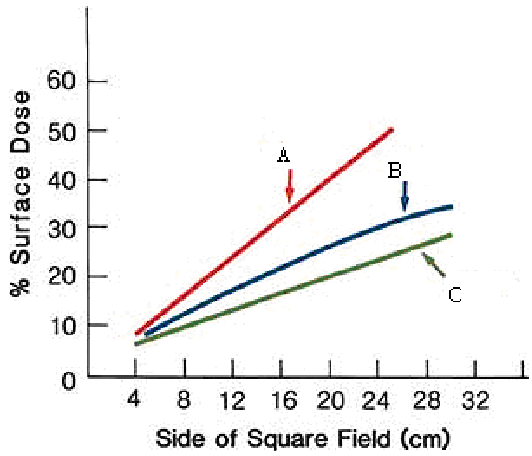
(A) a=increases, b=increases

(B) a=increases, b=decreases
(C) a=decreases, b=increases
(D) a=decreases, b=decreases

配合題：

27.The relative skin dose depends strongly on field size. Figure is a plot of relative surface dose as a function of field size for ^{60}Co, 4-MV, and 10-MV beams. Please to write down the correlation with A, B, and C.

A：___^{60}Co___ B:___4-MV___ C:___10-MV___



計算題：

28.A patient is treated with parallel opposed mantle and paraortic fields of lengths 30 and 15 cm, respectively. Calculate (a) the gap required on the surface for the beams to intersect at a midline depth of 10 cm and (b) the gap required to just eliminate the three-field overlap on the cord assumed to be at a depth of 15 cm from the anterior surface, given SSD = 100 cm for all the fields:

Ans:

$$S_1 = \frac{1}{2} \cdot L_1 \cdot \frac{d}{\text{SSD}} = \frac{1}{2} \times 30 \times \frac{10}{100} = 1.5\,\text{cm}$$

$$S_2 = \frac{1}{2} \cdot L_2 \cdot \frac{d}{\text{SSD}} = \frac{1}{2} \times 15 \times \frac{10}{100} = 0.75\,\text{cm}$$

Total gap required = 1.5 + 0.75 = 2.3 cm Ans: (a)

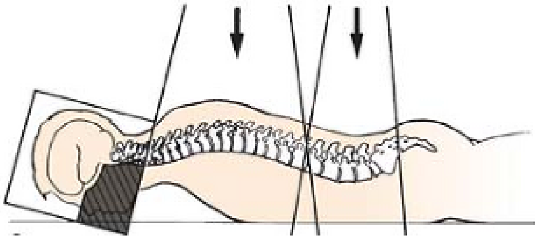
$$\Delta S' = \Delta S \cdot \frac{d' - d}{d} = 0.75 \times \frac{15 - 10}{10} = 0.4\,\text{cm}$$

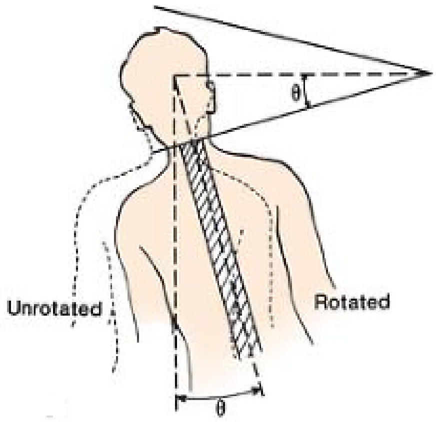
Length of three-field overlap on the cord:

$$\Delta S' = \Delta S \cdot \frac{d' - d}{d} = 0.75 \times \frac{15 - 10}{10} = 0.4\,\text{cm}$$

New gap required $= S_1 + S_2 + \Delta S' = 2.7$ cm Ans: (b)

29.For a patients is positioned prone and irradiated to cranial spinal for treatment of medulloblastoma, calculate the collimator rotation angle $\theta_{collimator}$ and couch rotation angles θ_{couch} to match diverging crainal fields with spinal field. The set up conditions are SAD=100 cm, Field size=20x20 cm^2 bilateral opposing for cranial fields and SSD=100cm, Field size=4x39 cm^2 posterior port for thorax spinal field.





Ans:

$\theta_{collimator} = \tan^{-1}(19.5/100) = 11.03^0$

$\theta_{couch} = \tan^{-1}(10/100) = 5.71^0$

A 30. Please select the ideal non-radiographic IGRT modality as the order listed below : good soft contrast, 3D real-time image tracking, real-time internal anatomy tracking, and abdominal soft-tissue guidance

(A) MRI, Optical Surface Tracking, RF Tracking, and Ultrasound
(B) MRI, RF tracking, Optical Surface tracking and Ultrasound
(C) Ultrasound, Optical Surface tracking, RF tracking and MRI
(D) Ultrasound, RF tracking, Optical Surface tracking and Ultrasound

C 31. Continuous tracking of patient motion during fixed gantry IMRT delivery can be

(A) CBCT with gantry –mounted kV source/detector

(B) CBCT with MV treatment beam

(C) Stereoscopic optical camera system

(D) All of the above

C 32. The benefits of optical surface imaging for breast patient setup as compared to portal films includes all of the following, **except :**

(A) No ionization radiation

(B) Real-time visualization of the body shape

(C) Better visualization of the lung/chest wall interface

(D) Provides 3D images

E 33. Which image-guided radiotherapy (IGRT) technique requires the least radiation dose:
(A) CBCT with gantry –mounted kV source/detector

(B) CBCT with MV treatment beam

(C) Planar kV radiographs

(D) Planar MV radiographs

(E) Radio-frequency (RF) beacon tracking

C 34. A disadvantage of using implanted radio-frequency beacons for patient setup and tracking is that beacons
(A) Require extra radiation dose

(B) Cannot be used for real-time patient tracking

(C) Produce artifacts on MR images

(D) Cannot be seen on planar x-Rays

E 35. Continuous tracking of prostate motion during fixed gantry IMRT delivery can be accomplished
(A) CBCT with gantry –mounted kV source/detector

(B) CBCT with MV treatment beam

(C) Stereoscopic optical camera system

(D) Stereoscopic infrared camera system

(E) Implanted RF electronmagnetic beacon

E 36. Which of these waves has the highest frequency?
(A) kV X-Ray

(B) Ultrasound

(C) Video tracking system

(D) Infrared Camera System

(E) Radiofrequency Tracking

C 37. The introduction of a Speckle Pattern is used to improve SNR for which system
(A) RF systems

(B) X-ray systems

(C) Visible light system

(D) Infrared Marker System

(E) Laser Scanning system

E 38. Which of the following is **<u>NOT true</u>** for RF location systems
(A) Near Real-time

(B) Can be used for gating

(C) Decreased dose to patient

(D) Accurate patient positioning

(E) Visualize internal anatomy

B 39. From TG 147: How frequently should localization accuracy be checked
(A) Before each patient

(B) Daily

(C) Weekly

(D) Monthly

(E) Yearly

B 40. From TG 147: What is the primary test used to check the localization
(A) Thermal Drift

(B) End to End Test

(C) Star Shots

(D) Calibration Test

(E) Latency Test

C 41. 現行高劑量率遙控後荷治療機，大多採用 ^{192}Ir 取代 ^{60}Co 或 ^{137}Cs，主要原因爲：
(A) 較長之半衰期
(B) 較高之光子能量
(C) 較高之比活性
(D) 較大之暴露率常數

C 42. Flethcr-Suit 裝療器，請問其 tandem 之曲度爲：
(A) 15°
(B) 15°,30°
(C) 15°,30°,45°
(D) 15°,30°,45°,60°

A　43. 使用 ^{125}I (Half life 60 days)射源植入攝護腺病人體內做永久性插種治療，若初始劑量率為 1cGy/h，則在一年後，病人接受多少的劑量？

(A) 20.7Gy

(B) 15.2Gy

(C) 10.1Gy

(D) 6.2Gy

B　44. 在近接治療中，對所使用的射源，其劑量率約定為？(Gy/hr)

(A) 0.4-5.0=LDR, HDR>5

(B) 0.4-2.0=LDR, HDR>12

(C) 1.0-3.0=LDR, HDR>20

(D) 1.0-5.0=LDR, HDR>12

B　45. 對於近接治療機中的 dummy source 與 radioactive source，兩者間的位置誤差應在多少以內？

(A) 0.5mm

(B) 1mm

(C) 1.5mm

(D) 2mm

C　46. 近接治療射源 Ra-226 由 0.5mm 的 Pt 過濾後，其暴露率常數為？R・cm^2・h^{-1}・mg^{-1}

(A) 6.25

(B) 7.25

(C) 8.25

(D) 9.25

D　47. 5.0Ci 等同於下列何者？

(A) 1.85MBq

(B) 7.4GBq

(C) 1.35・10^{-10}Bq

(D) 185GBq

B　48. 10.0Ci 經過 111 天後其活度約為多少？

(A) $9.25*10^{10}$Bq

(B) $1.31*10^{11}$Bq

(C) $1.85*10^{11}$Bq

(D) $2.47*10^{11}$Bq

C 49. 下列核種所放出的 r 射線能量最低？

(A) Cs-137

(B) Co-60

(C) I-125

(D) Ir-192

C 50. 下列插種系統，能對腫瘤治療的平面或體積，提供較均勻的劑量分佈？

(A) Quimby

(B) Memorial

(C) Manchester

(D) Paris